United International University
Department of Computer Science and Engineering
CSE 4811 / DS 4213: Natural Language Processing

Fall 2026

Practice Problems: Class 01

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1. **Types and tokens.**
   (a) What are the types and the tokens of a text?
   (b) Provide a sentence with exactly 4 types and 5 tokens. Explain any assumptions you make about the nature of tokens.
   (c) For the sentence "the children teased the dogs": how many word types are in it? For any word that is not already a lemma, write down its lemma.

   Cambridge *Machine Learning and Real-world Data* **2019** Paper 3, Q9(a)

   Edinburgh *Accelerated Natural Language Processing* **2025** Practice exam, Q3

2. **The size of the vocabulary.** Suppose you are training a unigram bag-of-words model. Which of the following preprocessing techniques will reduce the size of your feature space? List all that apply.

   I. Lowercasing the data
   II. Using bigram features instead of unigram features
   III. Stemming (removing suffixes from words, such as -ed from verbs or -s from plural nouns)
   IV. Removing stopwords (the, of, and so on)
   V. Tokenizing punctuation (splitting `great!` into `great !`)
   VI. Using tf-idf weighting

   UT Austin *CS371N Natural Language Processing* **2022** Midterm, Q1(5) · Solutions

3. **Building the bag.** Suppose you have the bag-of-words vocabulary [good, great, not]. Your training examples are simple but have some typos in them:

$$\begin{array}{ll} x = \text{goodx} & y = + \\ x = \text{great} & y = + \\ x = \text{not greatq} & y = - \\ x = \text{not good} & y = - \end{array}$$

   (a) Write down the feature vector for each of these examples, in the order of the vocabulary above.
   (b) Which examples can a bag-of-words classifier not tell apart from the empty text, and why?

   UT Austin *CS388 Natural Language Processing* **2023** Midterm, Q2(a) · Solutions

4. **Word order.** Suppose a bag-of-words model has been trained to decide whether the subject of a sentence is human or not. True or false, with a reason: it is possible that the model classifies both of the sentences "Francesco devoured the unsuspecting octopus" and "The unsuspecting octopus devoured Francesco" incorrectly.

   EPFL *CS-433 Machine Learning* **2024** Final, Q29 · Solutions

5. **How tf-idf behaves.** The tf-idf score for word w_i and document d_j ... (choose the one correct ending)

   (a) decreases with the number of documents in which w_i appears.
   (b) increases with the number of documents in which w_i appears.
   (c) is unaffected by the number of times w_i occurs in d_j.
   (d) increases with the count of w_i within documents $d_k \neq d_j$ in which it already appears.

   Toronto *CSC401 Natural Language Computing* **2017** Final, Q12

6. **Tf-idf vectors and their cosine.** Use the following raw counts for a few words in a collection of 4 documents, and assume that these are the only documents and words in the collection.

term	Doc1	Doc2	Doc3	Doc4
bicycle	36	30	0	3
tree	25	0	6	0
book	8	10	20	17
computer	1	38	35	0

Weight each term by $w = (1 + \log_{10} \text{tf}) \cdot \log_{10}(N/\text{df})$ when $\text{tf} > 0$, and 0 otherwise.

(a) Give the tf-idf vector of Doc1, in the order bicycle, tree, book, computer.

(b) Give the tf-idf vector of Doc3.

(c) Compute the cosine similarity between Doc1 and Doc3.

Stanford *CS124 From Languages to Information* **2019** Practice final, Q12 to Q14 · Solutions

7. **Why tf-idf.** Consider a standard bag-of-words model.

(a) Give an expression for the tf-idf weighting scheme which assigns a weight to each term in a document. Motivate each part of your expression.

(b) A new text is to be compared with the documents of the collection. Would you compute idf from the collection or from the new text? Why?

Cambridge *Information Retrieval* **2015** Paper 8, Q7(a)

8. **Dot product or cosine.** Consider the following vectors:

$$\text{good} = [0, 0, 3] \qquad \text{bad} = [0, 1, 3] \qquad \text{the} = [10, 10, 10]$$

(a) Report which two words are most similar using dot product similarity.

(b) Report which two words are most similar using cosine similarity, $\cos(a, b) = a \cdot b/(|a|\,|b|)$.

(c) Now suppose that all of the vectors are normalized to have length 1. Using the new vectors, report which two words are most similar using dot product similarity.

UT Austin *CS371N Natural Language Processing* **2022** Midterm, Q1(10) to (12) · Solutions

9. **Properties of cosine similarity.**

(a) What is the range of possible values for cosine similarity?

(b) Under what condition is cosine similarity equal to one? Show why it is equal to one in this case.

(c) Show that cosine similarity is invariant to rescaling one of its inputs. (Rescaling a vector means multiplying it by a positive constant.)

UMass Amherst *CS485 Natural Language Processing* **2024** Practice midterm, Q4.1, 4.2, 4.5

10. **Distance and cosine.** Let x and y be vectors of length one. Express the squared Euclidean distance $\|x - y\|^2$ through their cosine similarity. What does the result say about choosing the nearest vector by distance and by cosine?

UW *CSE 493G1 Deep Learning* **2023** Midterm, Q3.2 · Solutions

Edinburgh *Foundations of Natural Language Processing* **2024** Tutorial 5, Q2.4 · Solutions

11. **Ranking by cosine.** Given the query "elvis music" and the following term frequencies for the three documents doc1, doc2 and doc3, calculate the cosine similarity between the query and each document (you can ignore the idf term) in order to rank these documents in order of relevance. Show your workings.

	elvis	presley	mississippi	pop	music	life
doc1	3	4	0	6	0	0
doc2	4	0	4	0	0	3
doc3	5	3	0	4	4	0

Cambridge *Information Retrieval* **2018** Paper 8, Q7(b)(i)

12. **Each word its own axis.** The vector space model of document retrieval uses the notion of an information space.
    (a) What are the typical basis vectors for this space? What are the vectors in the space intended to represent?
    (b) How is term weighting used to position objects in the space? Why is term weighting important for effective document retrieval?
    (c) What is the orthogonality assumption typically employed in vector space models of document retrieval? Why might this assumption be false, and why might it lead to errors in retrieval?
    (d) Suggest one way in which the orthogonality assumption could be relaxed. (Just a short high-level description of a possible method is required.)

    Cambridge *Information Retrieval* **2010** Paper 8, Q9(a)

13. **From sentences to vectors.** Recall the definition of the cosine between two vectors x and y: $\cos(x, y) = \sum_i x_i y_i / \left(\sqrt{\sum_i x_i^2}\sqrt{\sum_i y_i^2}\right)$. How would you define a mapping from sentences s to vectors $f(s)$ such that $\cos(f(s_1), f(s_2))$ is a measure of the similarity between two sentences s_1 and s_2? Say what your measure captures and what it ignores.

    MIT *6.864 Advanced Natural Language Processing* **2006** Midterm, Q8